

llmXive Automated Review of Read It Back: Pretrained MLLMs Are Zero-Shot Reward Models for Text-to-Image Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper makes several specific quantitative claims in the Introduction and Experiments sections that do not align precisely with the reported results in the tables, creating ambiguity about the magnitude and conditions of the reported improvements.

First, the Introduction claims Self-`\method` outperforms the best external MLLM by “+2 on WISE.” However, Table 1 shows that without Chain-of-Thought (CoT), Self-`\method` (0.52) actually underperforms the best external MLLM (`\method`, 0.53). The gain of +0.02 (not +2) only appears when comparing the CoT variants (0.76 vs 0.74). The text fails to distinguish between the base method and the CoT-enhanced variant, leading to a significant numerical and methodological mismatch.

Second, the claim of “+4.3/5.5 on GenEval” in the Introduction is ambiguous. Table 1 shows a +5.5 gain over the BAGEL baseline (84.0 to 89.5) at 512 resolution. The source of the “4.3” figure is unclear from the table, as no other GenEval column or configuration yields this specific delta. This suggests a potential conflation of metrics or a missing specification of the comparison baseline.

Finally, the text asserts that Self-`\method` outperforms AlphaGRPO on GenEval at 1024 resolution. While the table lists Self-`\method` at 89.8, the corresponding AlphaGRPO entry for 1024 GenEval is missing (indicated by a dash). Without this data point, the claim of outperformance at this specific resolution is unsupported by the provided evidence.

These issues are primarily matters of precision and clarity in reporting results rather than fundamental flaws in the methodology. Correcting the text to accurately reflect the table data (e.g., specifying “with CoT” for the WISE gain, clarifying the GenEval deltas, and noting missing baselines) will resolve the discrepancies.

Action items.

- **[writing]** The paper makes several specific quantitative claims in the Introduction and Experiments sections that do not align precisely with the reported results in the tables, creating ambiguity about the magnitude and conditions of the reported improvements. First, the Introduction claims Self-`\method` outperforms the best external MLLM by “+2 on WISE.” However, Table 1 shows that without Chain-of-Thought (CoT), Self-`\method` (0.52) actually underperforms the best external MLLM (`\method`, 0.53). The gai

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear visual overview of the method and results, but the caption is severely truncated with missing method names and cut-off sentences, and panel (c) lacks a defined y-axis unit.

Figure 2

Figure 2 effectively communicates the architectural differences between four MLLM-based reward functions using clear block diagrams and concise text summaries. The visual layout aligns perfectly with the caption's description of each method's workflow and trade-offs.

Figure 3

The figure effectively visualizes token-level likelihood differences for positive and negative images, but the caption is marred by missing variable names and ambiguous phrasing regarding the data pairs.

Figure 4

The figure effectively demonstrates the correlation between reward scores and visual quality, but the caption is incomplete with missing method names and fails to define the numerical values shown on the images.

Figure 5

Figure 5 presents a visual comparison of image generation results but lacks a descriptive caption and internal labels, relying entirely on the viewer to infer the models and prompts from the images themselves.

Figure 6

Figure 6 effectively visualizes the reward ranking concept by displaying generated images ordered from lower to higher reward scores across four distinct prompts. The layout is clear, with numerical scores provided for each sample and a directional arrow indicating the reward gradient, fully supporting the caption's description.

Figure 7

The figure provides a clear qualitative comparison of image generation results against prompts, with red text effectively highlighting specific constraints. The layout is organized and the visual

evidence supports the caption’s claim that the Self-SpectraReward method satisfies constraints more consistently than the BAGEL baseline.

Action items.

- **[writing]** Figure 1: The caption text is truncated and contains placeholders (e.g., ‘Overview of .’, ‘aggregates this’, ‘Self-, using BAGEL’s own’) instead of the full method name ‘SpectraReward’, making it difficult to interpret the figure’s claims.
- **[writing]** Figure 1: The caption for panel (d) is cut off mid-sentence (‘bring significan’), failing to describe the content of the radar chart.
- **[science]** Figure 1: Panel (c) plots ‘Reward MLLM Comparison’ with a y-axis labeled ‘Reward’ but lacks a unit or scale definition (e.g., log-likelihood, normalized score), making the magnitude of the ‘non-monotonic gains’ uninterpretable.
- **[writing]** Figure 3: The caption contains multiple missing terms (e.g., ‘Token-level semantic sensitivity of .’, ‘aggregates this’, ‘i.e., .’) where variable names or method identifiers appear to be omitted.
- **[writing]** Figure 3: The caption text ‘calculated over four pairs’ is ambiguous; it is unclear if this refers to four image pairs or four token pairs, and the specific pairs are not defined.
- **[writing]** Figure 4: The caption contains a missing term (‘visual interpretation of .’) and the image lacks a label identifying the method (e.g., ‘Self-’) being visualized.
- **[writing]** Figure 4: The caption is extremely brief and does not explain the meaning of the numerical values (e.g., -3.490) displayed on the images.
- **[science]** Figure 5: The caption ‘Qualitative comparison’ is too generic and fails to describe the specific content shown (side-by-side image generation results for ‘BAGEL’ vs ‘Self-SpectraReward’ across four distinct prompts), making it impossible to understand the figure’s specific contribution without guessing.
- **[writing]** Figure 5: The figure contains no internal labels, axes, or legends to identify the specific prompts or the models being compared; all context is derived solely from the image content itself, which is poor practice for a standalone scientific figure.
- **[writing]** Figure 7: The caption contains a typo in the filename reference ‘[appendix_visualization_campare.pdf]’ (should be ‘compare’).

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-written and accessible to a competent researcher in adjacent fields

(e.g., computer vision or NLP). Most core concepts like “teacher-forced,” “log-likelihood,” and “diffusion models” are treated as standard vocabulary. However, there are a few instances of undefined acronyms and overloaded notation that create minor friction for a reader not deeply embedded in this specific subfield’s recent literature.

First, in Section 3.1, the variable T is used in Equation 1 to denote the number of prompt tokens. However, in Section 4.1, T is redefined as the number of sampling steps (“We use $T=16$ sampling steps”). While the context usually clarifies the meaning, using the same symbol for two distinct quantities in the same paper forces the reader to constantly check the section to disambiguate. Defining T explicitly as “prompt length” in Section 3.1 and using a different symbol (e.g., S or K) for sampling steps in Section 4.1 would eliminate this cognitive load.

Second, the acronym “AWM” is introduced in Section 4.1 as the default reinforcement learning algorithm without being spelled out. While it is a specific method, a reader from a neighboring field cannot infer what “AWM” stands for without external knowledge or a glossary. The full name should be provided at first use.

Third, in Section 4.1, the text mentions comparing against “AlphaGRPO... with the proposed MLLM-derived DVReward.” The term “DVReward” is an acronym that is never defined in the text. It appears to be a specific component of the cited AlphaGRPO work, but without a brief expansion (e.g., “Decomposed Verification Reward”), the reader is left guessing the nature of this baseline.

Finally, the term “teacher-forced” in Section 3.1 is standard in NLP but might be opaque to a pure vision researcher. A brief parenthetical explanation (e.g., “using ground-truth tokens as input”) would ensure the decoding strategy is clear to all readers.

Addressing these four points will make the paper fully self-contained for a competent adjacent-field PhD.

Action items.

- **[writing]** Section 3.1, Eq. 1: The symbol T is used in the denominator and summation limit without explicit definition in the text immediately preceding the equation. While defined later in Section 4.1 as ‘sampling steps’, in the method section it refers to prompt token count. Define T as the number of prompt tokens at its first use in Section 3.1 to avoid confusion with the sampling steps T used later.
- **[writing]** Section 3.1: The term ‘teacher-forced’ is used to describe the forward pass. While standard in NLP, a brief parenthetical clarification (e.g., ‘using ground-truth tokens as input’) would ensure a reader from a pure computer vision background understands the specific decoding strategy without needing to infer it from context.
- **[writing]** Section 4.1: The acronym ‘AWM’ is introduced as the default RL algorithm without being spelled out (e.g., ‘Adaptive Weighted...’). While it is a specific method name, defining the full name at first use in the Experimental Setup section is necessary for an adjacent-field reader to identify the algorithm.
- **[writing]** Section 4.1: The term ‘DVReward’ is used in the comparison with AlphaGRPO (‘...with the proposed MLLM-derived DVReward’) without definition. It appears to be a specific reward variant from the cited AlphaGRPO paper, but the acronym is not expanded

or explained in this text, leaving the reader to guess its meaning.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent argument for using image-conditioned prompt likelihood as a reward signal for text-to-image reinforcement learning. The core premise—that a pretrained MLLM’s ability to predict a prompt given an image reflects the image’s alignment with that prompt—is consistently defined and applied throughout the text.

The derivation of the reward function in Section 3 (Eq. 1) follows directly from the stated goal of measuring “how well the generated image can be translated back into the prompt.” The subsequent analysis of “language-prior cancellation” in Section 3.3 correctly identifies that for group-relative RL, the text-only prior cancels out, justifying the omission of a PMI-style correction without introducing a logical gap.

The distinction between the external \method and the internal Self-\method is maintained consistently. The claim that Self-\method benefits from “reward-policy alignment” is supported by the premise that the branches share tokenizers and pretraining distributions, and the experimental results in Table 1 and Table 2 consistently show Self-\method outperforming larger external models, which aligns with the hypothesis that alignment can rival scale.

There are no contradictions between the abstract, introduction, and conclusion regarding the scope of the results. The limitations section (Appendix) appropriately qualifies the method’s dependence on the MLLM’s visual reasoning capabilities, which is consistent with the method’s definition. The ablation studies (Tables 3 and 4) support the textual claims regarding the superiority of the likelihood-based reward over scalar scoring and VQA decomposition. The argument holds together without non-sequiturs or internal contradictions.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper generally maintains a strong alignment between its demonstrated results and its claims, particularly regarding the specific benchmarks and model families tested. However, there are three instances where the rhetoric slightly exceeds the strict scope of the evidence provided.

First, the abstract and introduction use universal quantifiers like “any pretrained MLLM” and “any” model. While the method is theoretically applicable to any MLLM, the empirical

evidence is strictly limited to four families (Qwen, Gemma, InternVL, BAGEL) and a specific scale range (4B to 235B). The paper does not test non-standard architectures or models outside this range. The claim should be hedged to reflect the specific diversity of the tested set rather than implying a universal guarantee across the entire landscape of MLLMs.

Second, the conclusion asserts that the method “significantly and consistently improves” performance across all benchmarks. A close reading of Table 4 (DPG-Bench) reveals that Self-\method does not outperform the external \method variant on the “Overall” score at 512 resolution (87.73 vs 88.08) and loses on the “Relation” category at 1024 resolution. The narrative treats the improvement as uniform, omitting these specific counter-examples. To maintain rigor, the text should acknowledge that while the trend is positive, consistency is not absolute across every metric and resolution.

Third, the paper introduces “Self-\method” as a “closed-loop self-improving framework.” While accurate for Unified Multimodal Models (UMMs) like BAGEL, the abstract’s broader framing could imply this loop is a general property of the method for all MLLMs. Standard MLLMs lack the distinct generation/understanding branches required for this specific self-rewarding loop without architectural changes. The conclusion should explicitly restrict the “self-improving framework” claim to the UMM setting to avoid overgeneralization to standard MLLMs.

These are primarily rhetorical refinements (severity: writing) rather than fundamental scientific flaws, as the core results are sound and the limitations are discussed in the appendix. Narrowing the universal claims and acknowledging specific metric failures will bring the paper’s rhetoric into perfect alignment with its evidence.

Action items.

- **[writing]** The abstract and introduction claim the method works for ‘any pretrained MLLM’ and ‘any’ model, but experiments are restricted to four specific families (Qwen, Gemma, InternVL, BAGEL) and a specific parameter range (4B-235B). Narrow the claim to ‘across diverse MLLM families tested’ or explicitly acknowledge the untested regimes (e.g., non-architectural variants, smaller/larger scales).
- **[writing]** The conclusion states the method ‘significantly and consistently improves... across all evaluated benchmarks,’ but Table 4 (DPG-Bench) shows Self-\method underperforming the \method variant on the ‘Overall’ score at 512 resolution (87.73 vs 88.08) and on ‘Relation’ at 1024 resolution. The narrative omits these specific failures. Add a sentence acknowledging that consistency is not absolute across all metrics and resolutions.
- **[writing]** The abstract claims the method is ‘training-free’ and ‘off-the-shelf,’ which is accurate for the reward function itself, but the conclusion frames the results as a ‘closed-loop self-improving framework’ without clarifying that this requires the specific architectural condition of a Unified Multimodal Model (UMM) with distinct understanding/generation branches. Generalize the ‘self-improving’ claim to apply only to UMMs, as standard MLLMs cannot form this loop without architectural modification.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a training-free reward mechanism for text-to-image generation using pre-trained Multimodal Large Language Models (MLLMs). From a safety and ethics perspective, the work is low-risk. The methodology relies on computing log-likelihoods of existing prompts against generated images, which does not inherently create new dual-use capabilities, nor does it lower the barrier to generating harmful content beyond what the underlying MLLMs and diffusion models already allow.

The authors have appropriately addressed the relevant ethical considerations in the “Broader Impacts” section (Appendix, `sec/X_suppl.tex`). They explicitly acknowledge that easier RL optimization could amplify existing risks, such as the generation of misleading visual content or unsafe images, and note the potential for inheriting biases from the reward backbone. They recommend standard safety filters and bias auditing as mitigations.

There are no indications of human-subjects data usage requiring IRB approval, no release of personally identifiable information (PII), and no use of scraped data in violation of terms of service (the paper relies on public benchmarks and pretrained models). The “Self-SpectraReward” approach, while creating a closed-loop self-improvement system, is a methodological contribution to alignment and does not constitute a system designed to deceive or surveil. The paper does not disclose operational vulnerabilities or provide actionable exploit code.

Consequently, there are no foreseeable, non-trivial risks of harm that the paper fails to acknowledge or mitigate. The existing disclosure is sufficient for this type of algorithmic research.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling hypothesis that image-conditioned prompt likelihood serves as an effective zero-shot reward signal. However, the experimental design contains specific gaps that prevent the reported results from definitively establishing the claimed causal mechanisms, particularly regarding the stability of the gains and the isolation of the “alignment” effect.

First, the main results in Tables 2 and 3 are presented as single-point estimates (e.g., Self-Method achieving 89.5 on GenEval) without any indication of variance. Reinforcement learning training is notoriously sensitive to random seeds, hyperparameter initialization, and stochastic sampling. A single run cannot distinguish a genuine improvement from a lucky seed or a specific trajectory. The absence of standard deviations, confidence intervals, or a stated number of seeds

(n) makes it impossible to assess whether the reported margins (e.g., +1.2 over the best external MLLM) are statistically significant or within the noise floor of the training process. To support the claim of “consistent improvement,” the authors must report results averaged over multiple independent training runs (at least 3-5 seeds).

Second, the comparison between Self-Method and external reward models (Table 2) conflates the quality of the reward signal with the computational cost and inference architecture. Self-Method leverages the policy’s own understanding branch, incurring zero additional inference overhead, whereas external MLLMs (like Qwen3-VL-30B) require separate, expensive forward passes. The paper attributes the superior performance of Self-Method to “reward-policy alignment,” but it does not rule out the alternative explanation that the external models were simply under-utilized or that the comparison is unfair due to the massive difference in inference cost. To isolate the “alignment” hypothesis, the authors should either report a compute-matched baseline (e.g., using a smaller external model that matches the inference cost of the self-reward) or explicitly demonstrate that the performance gap persists even when the external model is given equal computational resources or tuning effort.

Finally, the ablation study in Section 4.2 (Table 4) compares the proposed reward against scalar scoring and VQA-Score but does not fully control for the optimization algorithm and hyperparameters. The paper states that AWM was chosen because it performed best with Self-Method, but it does not show whether the same hyperparameters (learning rate, steps, group size) would yield similar gains if applied to a different reward function. It is plausible that the observed gains are partly due to the specific hyperparameter configuration being well-suited to the prompt-likelihood reward’s gradient properties, rather than the reward function itself being inherently superior. A more rigorous ablation would fix the hyperparameters across all reward types or report a grid search for each to ensure the comparison is fair.

Addressing these points—specifically by adding seed variance and controlling for compute/hyperparameter confounds—would significantly strengthen the evidentiary support for the paper’s central claims.

Action items.

- **[writing]** The paper presents a compelling hypothesis that image-conditioned prompt likelihood serves as an effective zero-shot reward signal. However, the experimental design contains specific gaps that prevent the reported results from definitively establishing the claimed causal mechanisms, particularly regarding the stability of the gains and the isolation of the “alignment” effect. First, the main results in Tables 2 and 3 are presented as single-point estimates (e.g., Self-Method achieving 89.5 on Ge

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment of the results in this paper is currently insufficient to support the strong claims of “significant” and “consistent” improvements. While the methodology for computing the reward (image-conditioned log-likelihood) is clearly defined, the inferential statistics applied to the benchmark results are missing.

First, the paper reports single-point estimates for all benchmark scores (e.g., GenEval, TIIF-Bench) in Tables 1 through 4 without any measure of uncertainty (standard deviation, standard error, or confidence intervals) or the number of random seeds used for training. In reinforcement learning, particularly with diffusion models, performance can fluctuate significantly due to stochasticity in the sampling process, initialization, and optimization trajectory. Reporting a single number (e.g., “89.5”) implies a level of precision and stability that is not justified without replication. The authors must report results as mean $\pm$ standard deviation over at least 3 independent training runs (seeds) to demonstrate that the observed gains are robust and not artifacts of a specific random seed.

Second, the paper makes frequent use of the term “significantly” (Abstract, Section 4.2) to describe the performance gaps between the proposed method and baselines. However, no hypothesis tests (e.g., t-tests, Wilcoxon signed-rank tests) are performed, no p-values are reported, and no corrections for multiple comparisons are applied. The study involves comparing the proposed method against multiple baselines (AlphaGRPO, various MLLM backbones, different RL algorithms) across five different benchmarks. This constitutes a large number of pairwise comparisons (likely >20). Without a correction for multiplicity (such as Bonferroni or Benjamini-Hochberg), the probability of observing at least one “significant” result by chance alone is high. The current reporting style treats the observed differences as definitive truths rather than estimates subject to sampling variance.

Finally, while Figure 2 mentions error bars calculated over “four pairs” for a specific token-level analysis, this level of uncertainty reporting is not extended to the main benchmark results. The discrepancy between the detailed error analysis in the figure and the point estimates in the tables creates an inconsistency in statistical rigor. To be scientifically sound, the uncertainty reporting must be consistent across all quantitative claims, and the claim of “statistical significance” must be backed by actual statistical testing or removed in favor of reporting observed effect sizes with confidence intervals.

Action items.

- **[science]** Tables 1-4 report single-point benchmark scores without uncertainty metrics (SD/SE/CI) or seed counts. RL results vary stochastically; report mean $\pm$ SD over 3 seeds for all main results to distinguish stable gains from noise.
- **[science]** Section 4.3 and Table 2 compare $\sim 20+$ configurations but claim ‘significant’ improvements without p-values or multiple-comparison corrections (e.g., Bonferroni). Apply FDR correction or rephrase claims to ‘observed improvements’ to avoid false positives.
- **[writing]** Figure 2 caption notes error bars over four pairs, but main tables lack uncertainty. Ensure consistent uncertainty reporting (e.g., SD over seeds) across all quantitative claims, not just in specific figures.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the core argument flows logically from the problem definition to the proposed solution and experimental validation. However, several paragraphs suffer from “buried main points,” where the primary claim or finding is placed after a list of details or specific examples, forcing the reader to infer the paragraph’s purpose until the end.

In Section 3 (Method), the “Analysis of \method” subsection contains a paragraph on “Token-level semantic sensitivity” that lists two types of mismatches (attribute and object identity) before explaining the experimental setup. The opening sentence is dense and mixes the general goal with specific instantiation details. A clearer structure would state the general investigation goal first, then describe the specific examples.

In Section 4 (Experiments), the “Main Results” paragraph opens with a procedural statement (“We use our \method...”) rather than the headline result. The paragraph then weaves together benchmark descriptions, baseline comparisons, and specific numerical gains. Reordering this to lead with the high-level finding—that the method consistently improves performance across all benchmarks—would significantly improve readability.

Similarly, in the “Effect of reward MLLM backbone” section, the paragraph regarding pretraining-stage MLLMs presents the specific result (Gemma3-12B-Pretrain outperforming the Instruct version) before offering the hypothesis explaining why. Moving the hypothesis to the start of the paragraph would provide necessary context for the data presented.

These issues are minor and do not obscure the scientific contribution, but addressing them would allow a reader to grasp the key points of each paragraph on the first pass.

Action items.

- **[writing]** Section 3, subsection ‘Analysis of \method’: The paragraph on ‘Token-level semantic sensitivity’ opens with a sentence that lists two failure types but then immediately pivots to describing the specific instantiation (counting error) without a clear transition. Split the first sentence to state the general finding (‘We examine sensitivity to local visual errors’) before detailing the specific examples of attribute and object mismatch.
- **[writing]** Section 4, subsection ‘Main Results’: The paragraph begins with ‘We use our \method as the proxy reward...’ but the subsequent sentences jump between benchmark descriptions, baseline comparisons, and specific numerical gains without a clear topic sentence. Restructure to first state the primary finding (e.g., ‘\method and Self-\method consistently outperform baselines across all five benchmarks’), then list the specific metrics and comparisons.
- **[writing]** Section 4, subsection ‘Effect of reward MLLM backbone’: The paragraph discussing ‘Pretraining-stage MLLMs’ starts with a specific result (‘Gemma3-12B-Pretrain consistently outperforms...’) before explaining the hypothesis. Move the hypothesis sentence (‘We hypoth-

esize that pretraining learns...') to the beginning of the paragraph to provide context for the observed result.